



HK IMO Training 2021-2022
Problem Set 26
For 23 April 2022



Problem 101. There are some clubs in a school. Every student can join any number of clubs they wish. It is known that every pair of clubs has at least one member in common. Prove that it is possible to put the students into two societies, such that at most one student belongs to both societies, and each society consists of students from every club.

Problem 102. In $\triangle ABC$, let D, E, F be the feet of altitudes from A, B, C respectively. Points P and Q are chosen on the line EF such that Q, F, E, P lie in this order, $EP = DF$ and $FQ = DE$. Point X is chosen on the line AB such that $XD = XP$. Point Y is chosen on the line AC such that $YD = YQ$. Prove that the line XY bisects BC .

Problem 103. Let $x_1 \geq x_2 \geq \cdots \geq x_n$ be positive real numbers. Prove that

$$x_1x_2(x_1 - x_2) + x_2x_3(x_2 - x_3) + \cdots + x_{n-1}x_n(x_{n-1} - x_n) \geq x_1x_n(x_1 - x_n).$$

Problem 104. For any positive integer n and any nonnegative integer $k \leq n$, let r_k be the remainder when $\binom{n}{k}$ is divided by 3. Find all positive integers n such that $r_0 + r_1 + \cdots + r_n \geq n$.